

A $r \times c$ matrix, $\bar{b} \in \mathbb{R}^r$, $\bar{x}$ unknown vector in $\mathbb{R}^c$

$Ax = b$: system of r linear equations in c variables

Augmented matrix $[A|b]$: each row corresponds to one equation, each column of A corresponds to a variable.

Small cases: 1 eqⁿ in 1 variable

$3x = 6$	1
$0x = 6$	0
$0x = 0$	∞

For now assume each eqⁿ is nonredundant and not inconsistent

# of vars	# equations	picture	# of solutions
2	1	line	∞ 1-dim
	2	intersection of two lines	$\infty, 0$ or 1
		same line parallel intersect in 1 pt	
	3	more cases. generic case?	generically 0 solutions
			
3	1	plane	∞ 2-dim
	2	intersection of two planes	$\infty, 0$ or ∞
		same parallel intersect in a line	better: 2d, 0, 1d

3 variables, 3 eq's $\leadsto$ get 3 planes A, B, C. Suppose no two same, no two parallel

Let $A \cap B = \text{line } L$

Solution set $= A \cap B \cap C = L \cap C$ 3 cases: $L \subset C$
 $L \parallel C$

generic case, $L \cap C = 1 \text{ pt}$

Solving $Ax=b$ by row reduction

- Solution set of $2x + 4y + 6z + 8w = 1$ is
 $(\Rightarrow x = \frac{1}{2}(1 - 4y - 6z - 8w))$

$$\left\{ \left(\frac{1}{2}(1 - 4y - 6z - 8w), y, z, w \right) \mid y, z, w \in \mathbb{R} \right\}$$

We call y, z, w free variables, x pivot variable

- 2 equations " $R_2 - 3R_1$ " means "Replace second row by second row $- 3(\text{first row})$ "

$$\begin{array}{rcl} 2x + 4y + 6z + 8w & = & 1 \\ 3x + \quad \quad 7z + w & = & 10 \end{array} \quad \left[\begin{array}{cccc|c} 2 & 4 & 6 & 8 & 1 \\ 3 & 0 & 7 & 1 & 10 \end{array} \right]$$

$$2 \text{ eq's } \uparrow \downarrow \frac{1}{2}(\text{eq } 1)$$

$$2R_1 \uparrow \downarrow \frac{1}{2}R_1$$

$$\Leftrightarrow \begin{array}{rcl} x + 2y + 3z + 4w & = & \frac{1}{2} \\ 3x + \quad \quad 7z + w & = & 10 \end{array} \quad \left[\begin{array}{cccc|c} 1 & 2 & 3 & 4 & 1/2 \\ 3 & 0 & 7 & 1 & 10 \end{array} \right]$$

$$\downarrow (\text{eq } 2) - (3 \text{ eq } 1)$$

$$R_2 + 3R_1 \uparrow \downarrow R_2 - 3R_1$$

you write

$$\left[\begin{array}{cccc|c} 1 & 2 & 3 & 4 & 1/2 \\ 0 & -6 & -2 & -11 & -17/2 \end{array} \right]$$

Rmk From this I can write down general sol'n as follows: ① take z, w arbitrary. ② use 2nd row/eqⁿ to solve for y in terms of z, w ③ use 1st eqⁿ to get

x in terms of y, z, w & substitute for y from step ②

We want to do better!

$-\frac{1}{6}R_2$

$$\left[\begin{array}{cccc|c} 1 & 0 & * & * & * \\ 0 & 1 & \frac{1}{3} & \frac{11}{6} & \frac{17}{12} \end{array} \right] \xrightarrow[\text{want to kill this 2}]{R_1 - 2R_2} \left[\begin{array}{cccc|c} 1 & 2 & 3 & 4 & \frac{1}{2} \\ 0 & 1 & \frac{1}{3} & \frac{11}{6} & \frac{17}{12} \end{array} \right]$$

Now I can solve (directly) for x & y in terms of z and w , which can take arbitrary values!

x, y : pivot variables

z, w : free variables $\rightarrow$ for now

solⁿ set has dim 2 $\stackrel{\text{def}}{=} \#$ of free variables

General procedure

① Reorder equations so that the leftmost nonzero entry is in the 1st row

② Scale 1st eqⁿ to make leading coeff 1

③ kill all entries below this leading 1

steps ① ② ③

• Continue the same procedure for matrix with rows 2 onwards

Notes added after class

- Leading nonzero entry in any row is called a pivot
- pivots move strictly to the right in successive rows

$$\left[\begin{array}{cccc|cccc} 0 & \dots & 0 & 1 & * & * & \dots & * \\ & & & 0 & 0 & 0 & 1 & * & \dots & * \\ & & & \vdots & & & 0 & & & \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & & \end{array} \right]$$

Shape of the result

- "We want to do better": can kill entries above each pivot too (entries below each pivot are already 0 as a result of previous work)

Questions to think about

- ① How will this method detect inconsistency of a system?
- ② Justify earlier heuristic observation for a consistent system of 4 eq^s in 10 variables: such a system must have a solution set of dimension at least 6